

Hands-On: Vision AI on the EK-RA8P1 with /IOTCONNECT



Hot-swap AI models onto an active microcontroller from the cloud and stream live video

60-minutes | Zack Andraka, Technical Solutions Engineer



Download/view this presentation deck at:

<https://github.com/avnet-iotconnect/iotc-freertos-ek-ra8p1/blob/main/docs/EK-RA8P1-vision-ai-workshop.pdf>

What We Will Achieve in this Workshop

FIND

Locate your device

Sign in, open the dashboard with your seat number, and watch your board's live telemetry arrive.

PUSH

Deploy AI Models

Deploy three different AI models from the cloud: a face detector, a 1000-class classifier, an occupancy sensor.

VALIDATE

Seamless Integration

Swap models while the board keeps running, then pull the power and see the model come back from flash.

WATCH

Stream Live Video

Open a live WebRTC video stream from your board on the dashboard and the device page, while the model keeps running.

Talk

Lab 1

Lab 2

Lab 3

Lab 4

Talk

Workspace Checklist

Board

- Pre-flashed and pre-configured EK-RA8P1
- Camera module connected
- LCD connected

Two Cables

- Ethernet
- USB-C

Unique Seat Card

- Seat number
- /IOTCONNECT Login (user+N) credentials

Your Laptop

- A browser open to awspoc.iotconnect.io
- A serial terminal installed (next slide)
- A free USB port for the board's cable

If your workspace is not as described above, please raise your hand to let an instructor know.

Before You Start: Open a Serial Terminal

BEFORE LAB 1

Lab 1 uses a serial terminal to show the board's console. The recommended one runs in your browser, so there is nothing to install.

Recommended: nothing to install

- **Chrome Labs Serial Terminal** runs in your browser on Windows, macOS and Linux
- Open googlechromelabs.github.io/serial-terminal in Chrome or Edge

Using Firefox or Safari?

- **Windows:** PuTTY (putty.sourceforge.net) or Tera Term (github.com/TeraTermProject/teraterm/releases)
- **macOS and Linux:** screen /dev/tty.usbmodem* 230400, or /dev/ttyACM0 on Linux

Lab 1 settings: 230400 baud, 8 data bits, no parity, 1 stop bit, no flow control.

Why an MCU for Vision AI?

Time Efficiency

- Boots in milliseconds
- Simple firmware reduces design and debug time
- Ethos-U55 completes face detection in **< 6ms**
- No worries of spending time on OS updates and re-designs for future incompatibilities

Power Efficiency

- Deterministic firmware on streamlined hardware results in minimal power needs
- CPU, RAM, and peripherals all on one chip so power isn't wasted sending data between components

Cost and Size Efficiency

- CPU, NPU, SRAM and flash in a single package
- No PMIC or high-speed DDR routing as an MPU design typically needs
- Fewer parts means a smaller, lower-cost board

BOTTOM LINE

MCUs are more than capable of supporting top-notch Vision AI applications, without the extra time and energy required with many MPUs.

How Does Model Deployment Work?

Here's what happens automatically after you click “Push Model” in /IOTCONNECT:

1

Send

a command from /IOTCONNECT to the board over MQTT with the model's download link.



2

Download

the model over HTTPS from S3 into a 4 MB holding buffer in SDRAM.



3

Validate

the file: a CRC checksum confirms it arrived intact, then a test load identifies the model type.



4

Hot-swap

the old model for the new one between two camera frames, reusing the same memory.



5

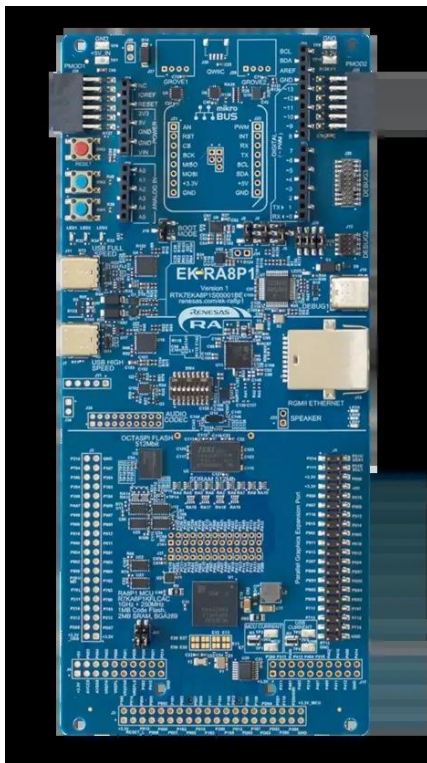
Save

the model to the board's flash storage so it survives a reboot or power loss.

FYI

- A hiccup at step 3 or 4 keeps the previous model running. The camera, display and telemetry never stop during any step.
- A push during a live video stream is deferred until the download can share the heap, then proceeds with the stream still playing.

The board: Renesas EK-RA8P1



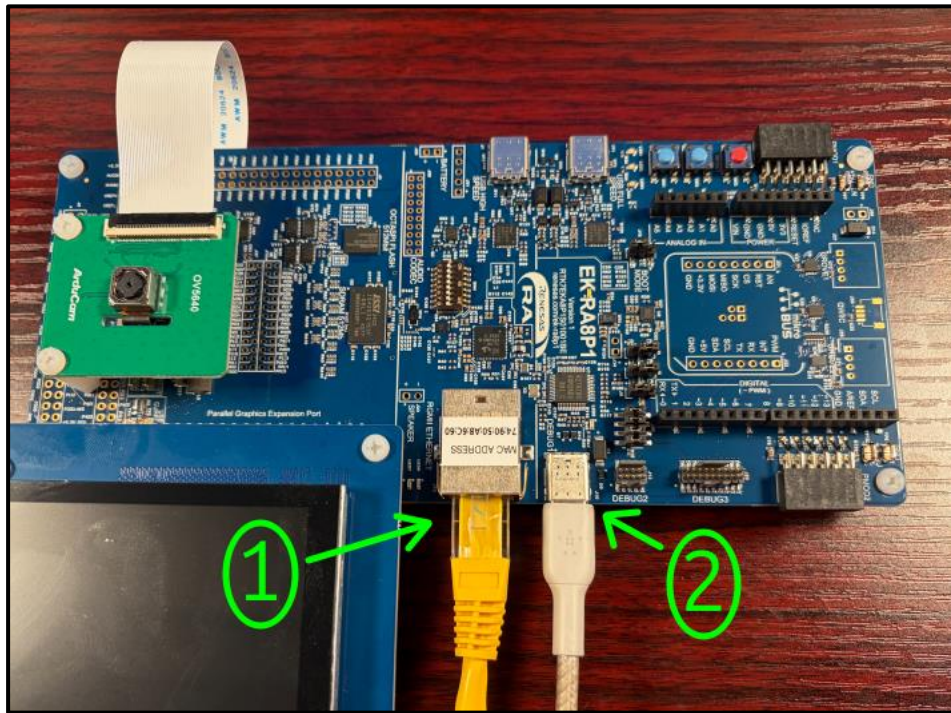
Part	What it is
MCU	RA8P1, Arm Cortex-M85 at 1 GHz
NPU	Arm Ethos-U55, 256 MACs per cycle
Memory	1 MB MRAM code, 2 MB SRAM, 64 MB SDRAM
Storage	64 MB OSPI flash: credentials and the model slot
Network	Gigabit Ethernet
Camera	OV5640 over MIPI CSI-2, 55 fps
Display	7-inch parallel LCD, optional
Debug	On-board J-Link; USB-C DEBUG1 is also the console

Lab 1

Open /IOTCONNECT Dashboard

Step 1: Hardware Connections

1. Plug the Ethernet cable into the board's Ethernet port.
2. Then plug the USB-C cable into the board's DEBUG1 port and a USB port on your laptop.



Step 2: Connect to the Serial Console

1. Open the **Chrome Labs Serial Terminal** in Chrome or Edge and enter the settings on the right.
2. Click **Connect** and choose the board's port (its name includes J-Link or USB Serial), then click **Connect** in the pop-up.
3. Within a few seconds the console shows a processing report every 5 seconds (camera, AI and display timing) and lines starting with IOTC: confirming the board is connected.

SERIAL SETTINGS

Baud rate	230400
Data bits	8
Parity	None
Stop bits	1
Flow control	Off (unticked)
Convert EOL	On (ticked)

WHAT YOU SHOULD SEE

```
Processing time:
  Camera image capture vsync period : 18 ms, 55 fps
  AI inference time (Ethos-U55)      : 0 us, 0 fps
IOTC: hb up=95 netbuf=16 heap=181360 conn=1 sent=6
```

If you see no output, or the text is garbled, please raise your hand for assistance.

Step 3: /IOTCONNECT Login

1. In your browser, go to **awspoc.iotconnect.io**.
2. Log in to /IOTCONNECT with the username and password on your seat card.

 awspoc.iotconnect.io

Welcome back,

Please login to your account

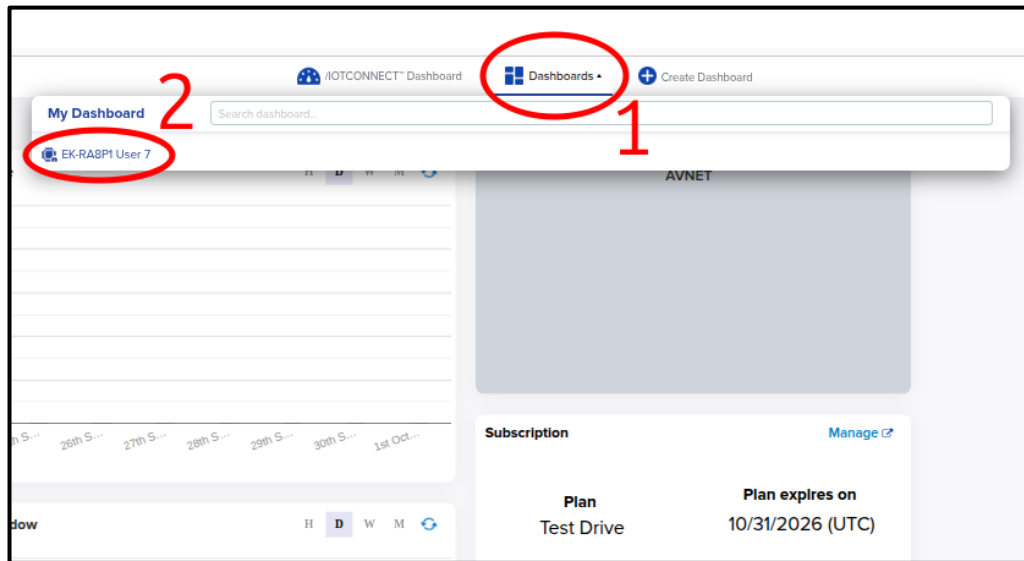
 iotc-user+7@gmail.com

[Forgot Password?](#)

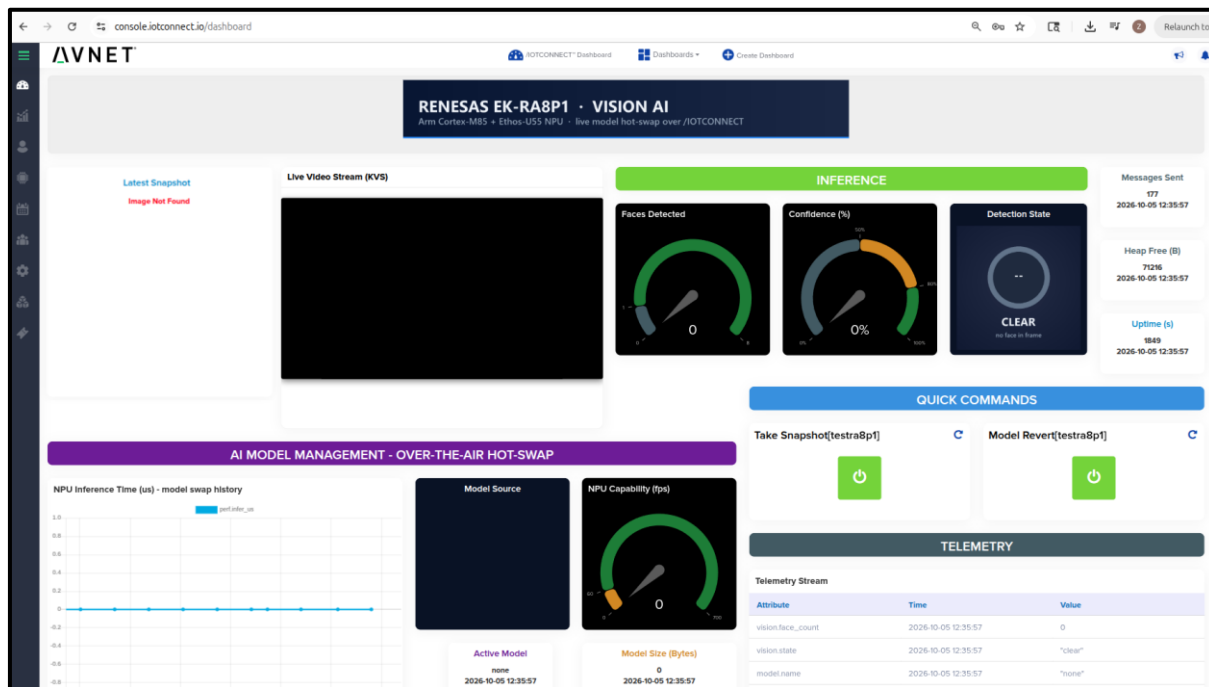
Next

Step 4: Open the /IOTCONNECT Dashboard

1. Click **Dashboards** in the toolbar at the top of /IOTCONNECT.
2. Open the dashboard titled "EK-RA8P1 User N", where N is the user number on your seat card.



Expected Dashboard State



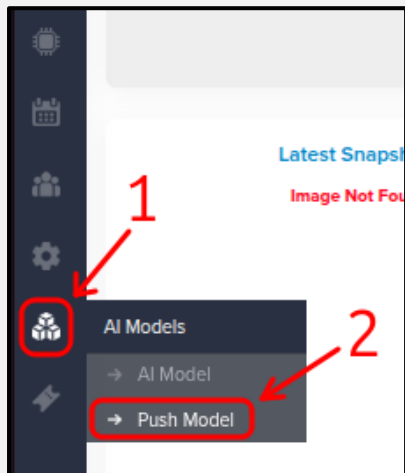
If your dashboard does not look like this, please raise your hand for assistance.

Lab 2

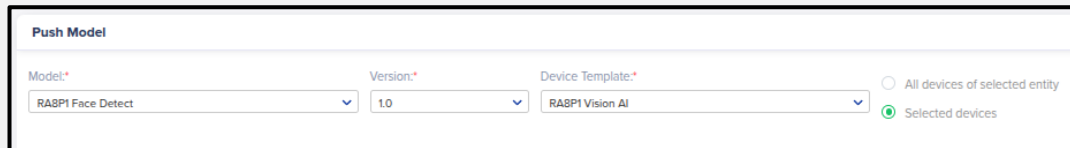
First AI Model Deployment

Step 1: Deploy the Face Detect Model

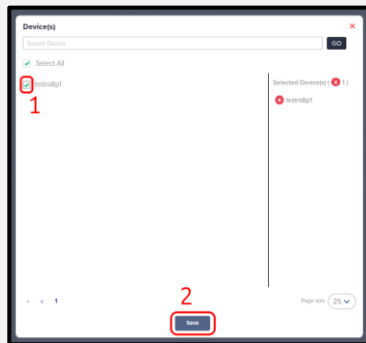
1. Hover over the blocks icon in the left-hand toolbar, then click **Push Model**.



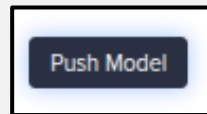
2. Select N - RA8P1 Face Detect for **Model** (Version is 1.0), RA8P1 Vision AI for **Device Template**, then click **Selected devices**.

A screenshot of the 'Push Model' dialog box. It contains three dropdown menus: 'Model' (set to 'RA8P1 Face Detect'), 'Version' (set to '1.0'), and 'Device Template' (set to 'RA8P1 Vision AI'). There are two radio buttons: 'All devices of selected entity' (unselected) and 'Selected devices' (selected). The 'Selected devices' radio button is highlighted with a green circle.

3. Click **Select Device**, check the box next to your device's unique ID, then click Save.



4. Click the **Push Model** button.



Tip: model names start with your user number N.
User 5 picks **5 - RA8P1 Face Detect**

Step 2: Verify the Model Hot-Swap

1. In your serial terminal, the push arrives within a second or two:

```
IOTC: model push from https://iotc-575198115013.s3.us-east-1.amazonaws.com/c0327ced-5af4-4a06-ba26-55d328d2252e/model/AI%20Model/Renesas/dbabdf7f-0030-422f-87a
```

2. Over the next 10–30 seconds the board downloads the model, checks it and swaps it in. It is ready when the last line reads FD: model "face-v3" v3 ... loaded.

```
IOTC: model downloaded (441248 bytes)
IOTC: unpacked STORED zip -> 441120 bytes
FD: hot-swapping to model "face-v3" v3 (441088 bytes)
INFO - Added ethos-ai support to sp resolver
INFO - Creating allocator using tensor arena at 0x02209e410
INFO - Allocating tensors
INFO - Model INPUT tensors:
INFO - tensor type is INT8
INFO - tensor occupies 36864 bytes with dimensions
INFO - 0: 1
INFO - 1: 192
INFO - 2: 192
INFO - 3: 1
INFO - Quant dimension: 0
INFO - Scale[0] = 0.009321
INFO - ZeroPoint[0] = -428
INFO - Model OUTPUT tensors:
INFO - tensor type is INT8
INFO - tensor occupies 640 bytes with dimensions
INFO - 0: 1
INFO - 1: 6
INFO - 2: 6
INFO - 3: 18
INFO - Quant dimension: 0
INFO - Scale[0] = 0.124084
INFO - ZeroPoint[0] = 47
INFO - tensor type is INT8
INFO - tensor occupies 2592 bytes with dimensions
INFO - 0: 1
INFO - 1: 42
INFO - 2: 42
INFO - 3: 18
INFO - Quant dimension: 0
INFO - Scale[0] = 0.105359
INFO - ZeroPoint[0] = 10
INFO - Activation buffer (a.k.a tensor arena) size used: 443864
INFO - Number of operators: 1
INFO - Operator 0: ethos-ai
FD: model "face-v3" v3 (cloud, 441088 bytes) loaded: face detector, ethos-ai: yes
```

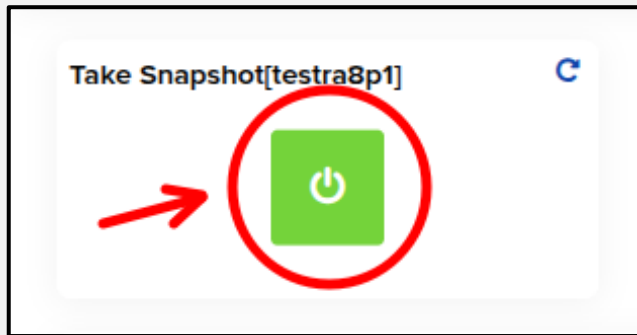
3. On the LCD, the Model line changes from none v0 to face-v3 v3.

```
Model: none v0
↓
Model: face-v3 v3
```

4. Position your face in front of the camera. The board should recognize your face right away and draw a green bounding box around it.

Step 3: Capture a Snapshot from the Dashboard

1. Go back to your /IOTCONNECT dashboard.
2. Position your face in front of the camera and press the green **Take Snapshot** button near the bottom right of the dashboard.



3. Within about 30 seconds the board uploads the photo to /IOTCONNECT, and the **Latest Snapshot** widget shows it with the green box drawn around your face.



Lab 3

Deploying and Hot-Swapping More Models

Step 1: Deploy an Image Classification Model

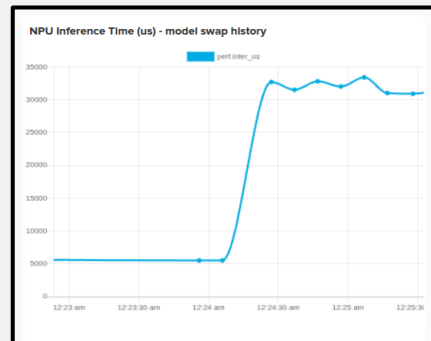
1. Return to the Push Model page and use the same settings as before, but choose N - RA8P1 ImageNet Classifier v2 for Model.

3. When the swap completes, the Model line on the LCD changes to mobilenet-v2 v1.

```
Model: face-v3 v3  
↓  
Model: mobilenet-v2 v1
```

2. This model is about 7× the size of the face detector, so the download and swap take one to two minutes. Watch the serial console for the model info stream.

4. Back on your dashboard, the NPU Inference Time chart (bottom left) jumps from about 6 ms to about 40 ms as the larger model takes over.



Step 2: Deploy the Person Detect Model

1. On the Push Model page, use the same settings again and choose N - RA8P1 Person Detect. It is the smallest model, so it arrives much faster.

3. Hold the board out with the camera facing you: Class changes from no person to person. Score shows the model's confidence, and inference takes about 1.5 ms.

```
Class: no person
↓
Class: person
Score: 95%
```

2. After the swap, the Model line on the LCD changes to person-detec v2.

```
Model: mobilenet-v2 v1
↓
Model: person-detec v2
```

4. On your dashboard, the same NPU Inference Time chart drops to about 1.5 ms.

Step 3: Validate Model Persists Upon Reboot

1. On the LCD, Source reads cloud: the model came from /IOTCONNECT and was plugged straight into the running application.

```
Source: cloud
```

2. Unplug the USB-C cable from the board, wait 3 seconds, then plug it back in.

3. When the LCD comes back, Source reads flash and Model still reads person-detec v2.

```
Model: person-detec v2  
Source: flash
```

The model was hot-swapped into the running application and also saved to non-volatile flash, so it stays active through a power cycle.

Lab 4

Live Video with KVS WebRTC

Why Watch Live Video?

SEE IT FOR YOURSELF

Alerts tell you what the model detected. Live video lets you see it yourself, to validate the inference or decide your next step.

Low Latency

- Peer-to-peer WebRTC: video goes straight from the board to your browser
- No cloud recording or transcoding in the path
- Video appears within seconds of clicking Start

Secure Anywhere

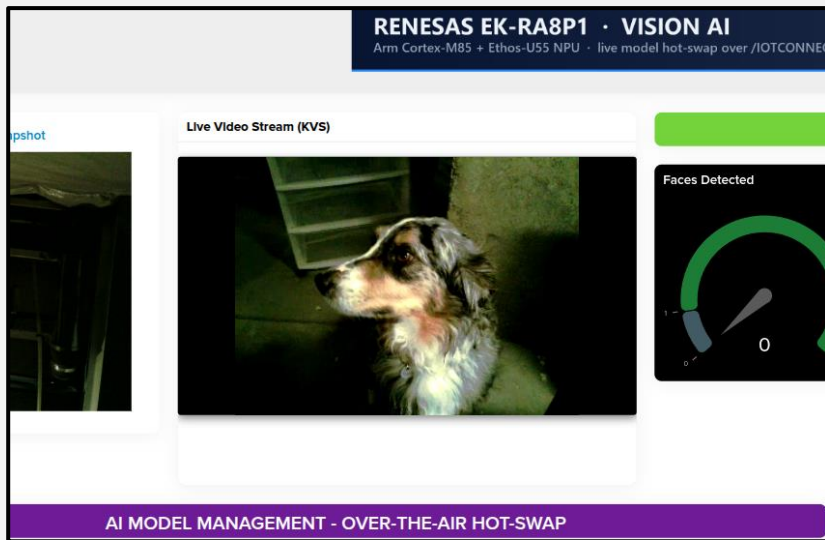
- Encrypted end to end with **DTLS-SRTP**
- ICE, STUN and TURN get through NAT and firewalls, with no port forwarding
- Signaling channel created automatically by /IOTCONNECT

Efficient on an MCU

- H.264 encoded in software on the Cortex-M85: **320×240 at 8–10 fps, ~500 kbit/s**
- Encodes only while someone is watching
- Inference and telemetry keep running during the stream

Step 1: Preview the Stream on Your Dashboard

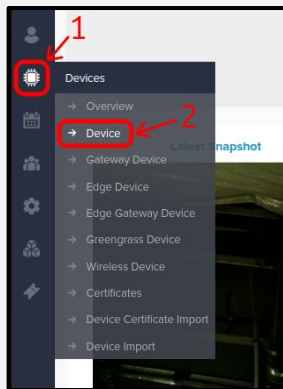
1. The quickest way to see the video is the Live Video Stream (KVS) widget on the dashboard you opened in Lab 1.
2. Use it for a quick glance: when telemetry or an alert reports a detection, check the preview to confirm what the model saw, from anywhere.



For a larger view of the same stream, open it from the device page in the next step.

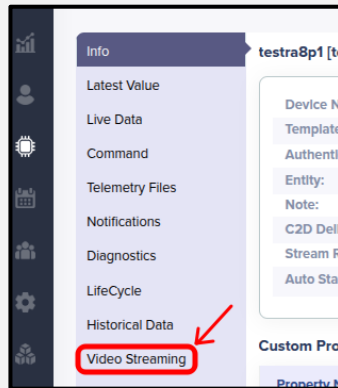
Step 2: Full-Screen View on the Device Page

1. Hover over the processor icon in the left toolbar, then click Device.

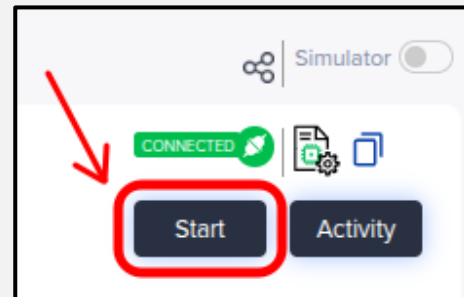


2. Click your device's unique ID to open its device page.

3. Click the Video Streaming tab.



4. Click Start in the top-right corner. The same live stream opens in a large, nearly full-screen player.



“Start” opens a peer-to-peer WebRTC connection and becomes “Stop,” which ends the stream.

The provided firmware only supports 1 viewer connection, so you cannot have the dashboard and Video Streaming both open simultaneously.

AI Model Catalog

In AI Models	What the board becomes	Input	Inference	Size
RA8P1 Face Detect	Face detector with boxes (YOLO Fastest)	192×192 gray	5.8 ms	441 KB
RA8P1 Person Detect	Occupancy sensor: person present or absent	96×96 gray	1.5 ms	270 KB
RA8P1 ImageNet Classifier 0.25	1000-class classifier, speed tier	224×224 RGB	4.5 ms	432 KB
RA8P1 ImageNet Classifier 0.5	1000-class classifier, mid tier	224×224 RGB	9 ms	1.1 MB
RA8P1 ImageNet Classifier v2	1000-class classifier, accuracy tier	224×224 RGB	40 ms	3.1 MB

Camera Max FPS: 55 (new frame every 18ms)

Slowest Model FPS: 25

Significance: Provided models never need more than ~2 frames to make their inferences.

Classifier v2 is roughly seven times the face detector's size and workload. That is the one to push while you watch the inference chart.

Key Takeaways

AI models are cloud-managed assets

Deploy and update a model from /IOTCONNECT like any other data, to one board or a whole fleet, with no firmware rebuild or reflash.

Vision AI belongs on the MCU

The Cortex-M85 and Ethos-U55 run detection and classification in milliseconds, on one low-cost, low-power chip.

Verify from anywhere

Telemetry, snapshots and live WebRTC video let you check what the device saw and decide the next step remotely.

REAL-WORLD EXAMPLE

Smart Building Occupancy

- **Detect:** A camera node in every meeting room runs person detection on its MCU and reports occupancy to /IOTCONNECT, driving lighting, HVAC and room booking.
- **Update:** When a better model is trained, facilities pushes it to every node in the building overnight, with no site visits.
- **Verify:** A room reports occupied after hours. The manager opens the live video to check before sending security

Each lab today built one piece of it.

Next Steps

Build your own model and keep exploring

Create and Deploy Your Own Model

1

Train and quantize

Export an int8 TensorFlow Lite model whose input matches a supported model family.

**2**

Compile with Vela

Target ethos-u55-256. If Total SRAM used exceeds 640 KiB, recompile with the Size optimisation.

**3**

Pack

tools/pack_model.py wraps it in the IOTV envelope and writes an uncompressed zip.

**4**

Upload and push

In AI Models, create a model with Variant Renesas and upload the zip. Push it to one device, then the fleet.

Supported model families

- Face detector: 192×192 grayscale input (YOLO Fastest)
- Classifier: 32–224 px gray or RGB input, with 2 or 1000 classes

Limits

- Up to 4 MB packed and 640 KiB of Vela SRAM
- Must run entirely on the Ethos-U55 NPU
- An unsupported model is rejected, and the previous one is restored

Exact commands and details are in the Developer Guide, section 9: AI Models.

Renesas Examples on /IOTCONNECT

Example	What It Shows	Boards	Repo
EK-RA8P1 Vision AI Hot-Swap	This workshop; cloud model hot-swap on NPU	EK-RA8P1	iotc-freertos-ek-ra8p1
RZ/V2H AI Inference Vision	Python vision AI inference pipeline	RZ/V2H EVK	iotc-python-lite-sdk-demos
RZ/V2H System Monitor	Streams live system metrics	RZ/V2H EVK	iotc-python-lite-sdk-demos
Managing Edge AI: RZBoard V2L	On-board AI camera quickstart	RZBoard V2L	meta-iotconnect-docs
SPARK Smart Parking	CNN parking and EV spot detection	RZBoard V2L	SPARK
RASyn Puck	Audio edge AI out-of-box demo	RASynBoard NDP120	RASynBoard-Out-of-Box-Demo
RA Cloud Ready Platforms	Wi-Fi telemetry and LED commands	EK-RA8M1, EK-RA6M4	iotc-freertos-ek-ra8m1-pmod
Machine Monitoring	Predictive maintenance over Wi-Fi	CK-RA6M5	iotc-freertos-ck-ra6m5-v2-pmod
EK-RA6M4 PMOD Gateway	FreeRTOS sample with DA16600 Wi-Fi/BLE	EK-RA6M4	iotc-freertos-ek-ra6m4-pmod
FreeRTOS on EK-RA6W1	Built-in Wi-Fi 6, secure MQTT/TLS	EK-RA6W1	iotc-freertos-ek-ra6w1
LoRaWAN Sensor Monitoring	LoRaWAN end nodes with HS3001 sensors	EK-RA2L1, RA FPBs	iotc-lora-demos
C SDK for Azure RTOS	SDK and sample apps for Azure RTOS	CK-RX65N	iotc-azurerτος-sdk
DA16K SDK	SDK for DA16K Wi-Fi modules	DA16200, DA16600	iotc-dialog-da16k-sdk
DA16K AT-Command Library	FreeRTOS AT-command C library	DA16K PMOD, RA MCUs	iotc-freertos-da16k-atcmd-lib
RZ/G3E Python Lite SDK	Python telemetry, commands and OTA	RZ/G3E EVK	iotc-python-lite-sdk-demos

FULL CATALOG avnet-iotconnect.github.io/iotc-index select Renesas under Partners

Explore the Repository

THE REPOSITORY

github.com/avnet-iotconnect/iotc-freertos-ek-ra8p1

Quickstart

- Repeat today at home
- Prebuilt image, no toolchain
- From unboxing to live video

Developer Guide

- Build from source with e² studio
- Architecture and memory budget
- Adding your own models

Also in the repo

- Prebuilt firmware image
- Device template and dashboard
- Five ready-to-push models
- These workshop slides

MORE RENESAS /IOTCONNECT DEMOS

CK-RA6M5 Cloud Kit

Live sensor data over a DA16K Wi-Fi PMOD.

github.com/avnet-iotconnect/iotc-freertos-ck-ra6m5-v2-pmod

EK-RA6W1 Wi-Fi MCU

Telemetry and commands over the RA6W1's built-in Wi-Fi.

github.com/avnet-iotconnect/iotc-freertos-ek-ra6w1

All /IOTCONNECT Projects

SDKs, demos and documentation.
github.com/avnet-iotconnect

Thank you
